

AVAW_CV-Waste: An Offline Computer Vision Annotation Tool for Waste Management Research

Gerald Koinig^{1¶} and Bojan Lorber¹

¹ Montanuniversität Leoben, Austria ¶ Corresponding author

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1 Summary

AVAW_CV-Waste is a free and open-source offline desktop application for creating and correcting image annotations used to train object detection models. The software allows users to draw, edit, move, resize, copy, and manage bounding boxes around objects in images. It creates outputs compatible with YOLO-based artificial intelligence workflows, specifically the YOLO PyTorch TXT annotation format commonly used by YOLO11, YOLOv8, YOLOv5, and related computer vision models.

AVAW_CV-Waste supports the integration of existing YOLO models for semi-automatic annotation generation, reducing the amount of manual labelling required during dataset preparation. This assistance can be used to annotate complete images or follow a one-click annotation process in which the user clicks on an object and the currently loaded models tries to fit a bounding box around the selected object. The application was designed specifically for waste management, recycling, and material sorting research, where datasets often contain highly domain-specific object classes and cannot easily be processed using generic or cloud-based annotation platforms.

Because the software operates entirely offline, it can be deployed directly at industrial facilities or research sites where internet access may be limited or unavailable. Local processing additionally guarantees data sovereignty, an important requirement for industrial research collaborations involving sensitive operational data.

The primary goal of AVAW_CV-Waste is to simplify and accelerate dataset creation for non-specialist users, including students, laboratory staff, recycling operators, and industrial research partners. By reducing the technical complexity of annotation workflows, the software supports the development of custom machine learning models for applications such as waste classification, contaminant detection, scrap sorting, and sensor-based recycling research.

2 Statement of need

AVAW_CV-Waste was developed as a free and open-source alternative that can be adapted to specific scientific workflows, especially in waste management and recycling research.

The Waste Management domain needs images of particles with contamination, deformation and on conveyor belts, specific to the actual research question regarding the sorting and classification of waste particles are necessary. Thus, waste datasets often require highly specific object classes, such as plastics, paper, scrap-types, hazardous waste, or mixed waste fractions in environments relevant to the waste management domain. Publicly available datasets that fit all these requirements are scarce to non-existent and thus need to be created for each research project individually.

Existing annotation tools for this purpose may not easily support these specialized categories

40 or the frequent changes that occur during research projects and often prohibit or disincentivise
41 the use of custom pre-trained models, which may impede up the annotation process.

42 Lastly, many existing tools are cloud based, and thus data sovereignty may not always be
43 guaranteed. This is an increasingly precarious issue when working with industrial partners in
44 general and partners in waste management in particular due to research projects requiring data
45 to be recorded directly in respective material recovery facilities.

46 This tool allows users to define their own custom annotation classes, load and use their own
47 models for automatic predictions and to work entirely offline without requiring cloud services.

48 The software primarily targets researchers and industrial practitioners working in waste
49 management, recycling, circular economy technologies, and sensor-based sorting applications
50 enabling them to quickly create, review, edit, and correct annotations at no cost.

51 **3 State of the field**

52 The field of annotation software allows for several pre-build alternatives.

53 However, these tools' reliance on online annotation with the requirement to upload the dataset
54 and related annotations into a cloud service often clashes with industrial partners need for
55 data sovereignty. Additionally, existing annotation tools may not easily integrate with the
56 specialized categories or the frequent changes that occur during research projects.

57 In addition, many annotation tools restrict or disincentivise the integration of custom pre-
58 trained models, thereby limiting opportunities to accelerate the annotation process through
59 domain-specific automation. While some cloud-based services provide server-side automated
60 annotation using existing pre-trained models, such functionality is often confined to paid
61 subscription tiers and restricted to the models supported by the respective platform, or simply
62 limited to set number of uses if using free subscription tiers. Consequently, researchers are
63 frequently unable to leverage their own waste-management-specific pre-trained models to
64 support and aid annotation.

65 In principle, many of these limitations can be circumvented by deploying self-hosted instances
66 of existing annotation platforms on offline systems. However, accomplishing this typically
67 requires advanced technical expertise, and, most importantly, a considerable time investment.

68 Consequently, the emerging research field of employing machine learning models for waste
69 classification tasks requires an open source, free of cost, offline and adaptable tool that
70 allows for modular expansions and use of domain specific detection models that also ensures
71 data-sovereignty and is not reliant on online services once installed.

72 AVAW_CV-Waste addresses this gap by occupying a niche between generic cloud-based annotation
73 platforms and highly specialized in-house software solutions commonly employed in industrial
74 recycling research.

75 **4 Software design**

76 While generative AI can now accelerate the initial development of software like this, extensive
77 testing in real research environments is still required. In particular, evaluating the tool with
78 researchers of different technical backgrounds cannot be meaningfully outsourced or offloaded
79 to generative AI systems.

80 Thus, the UI and UX of this tool were subject to constant iteration and improvement, spanning
81 multiple years of research projects in this emerging field in waste management research. It
82 could therefore incorporate criticism and feedback from students during lectures, as well as
83 from researchers and workers that annotated thousands of images using this tool. Ongoing use
84 and the evolving requirements of research projects, particularly in waste-management-related

85 sorting tasks, will inevitably uncover further opportunities for optimisation and adaptation.
86 The use of Python, the widely adopted UI framework Tkinter, and an object-oriented software
87 architecture is intended to support straightforward customisation and extension by researchers
88 in waste management and recycling domains.

89 To maintain simplicity and accessibility, the software intentionally avoids implementation
90 complexity, such as multithreading or multitasking even though these approaches could have
91 potentially improved application responsiveness by offloading annotation tasks to separate
92 processes.

93 Since Python is already widely used in machine learning and computer vision research, the
94 tool integrates naturally into existing scientific workflows.

95 Several design trade-offs were made to ensure broad hardware compatibility and stable operation
96 on lower-performance research machines while still providing the quality-of-life features expected
97 from such a tool. Lightweight rendering and simplified interaction workflows were prioritized
98 over computationally expensive visualization features. Although the default Python environment
99 is primarily designed for CPU-based inference, the underlying inference modules also enable
100 advanced users to leverage available GPU hardware resources with only minor changes to the
101 installation workflow.

102 Overall, Python and its ecosystem of modules were selected as the underlying architecture
103 due to the broad support available for Python applications, the ease with which other waste
104 management researchers can adapt the tool to their own requirements, and the widespread
105 adoption of Python for the rapid development of machine learning applications.

106 5 Research impact statement

107 AVAW_CV-Waste has already been successfully used in the development of scientific datasets
108 and in research related to waste detection, classification, and recycling workflows.

109 AVAW_CV-Waste addresses a major bottleneck in the development of computer vision systems
110 for waste management and recycling research, namely the creation of annotated training
111 datasets.

112 AVAW_CV-Waste is currently being used within multiple recycling and circular economy lighthouse
113 research projects, including KiRAMET, StraTex, greenPLAST-food, and Scarpa.

114 Within these projects, the software has been applied in research workflows involving post-
115 consumer textiles, lightweight packaging waste, and post-shredder scrap, where large quantities
116 of annotated image data are required for training and evaluating object detection models.

117 Datasets created using AVAW_CV-Waste contributed to research on sensor-based sorting of
118 shredder scrap towards green steel production ("[AI Pipeline for Sensor Based Sorting of
119 Shredder Scrap Towards Green Steel Production](#)," 2026), AI-assisted recycling workflows
120 for metal composite waste streams ([KiRAMET](#), 2026; [Kiramet](#), 2026), and plastic recycling
121 systems for food-contact materials ([GreenPLAST-Food](#), 2026).

122 The generated datasets also supported research on robust YOLO-based ejection of copper-
123 containing particles from heavily corroded scrap streams ("[Robust YOLO-Based Ejection of
124 Copper-Containing Particles in Heavily Corroded Scrap Towards Green-Steel Production](#),"
125 2026), detection of copper-containing scrap in post-shredder fractions using machine vision and
126 artificial intelligence ([Detection of Copper-Containing Scrap in a Post-Shredder Fraction with
127 Machine Vision and Artificial Intelligence Towards Green-Steel Production](#), 2026), and deep
128 learning approaches for classification of copper-containing metal scrap in recycling processes
129 ([Deep Learning Approaches for Classification of Copper-Containing Metal Scrap in Recycling
130 Processes](#), 2026).

131 These applications demonstrate the utility of the software for reducing manual annotation
132 effort and enabling machine learning methods for heterogeneous waste sorting tasks.

133 AI usage disclosure

134 Generative artificial intelligence tools were not used for the architectural design or core software
135 implementation of AVAW_CV-Waste.

136 ChatGPT (OpenAI) was used during software development for documentation support,
137 generation of docstrings, bug-fixing assistance, and inline code modifications. For this
138 manuscript, generative AI tools were used only for grammatical improvements and formatting
139 support, including conversion of manuscript content from .docx to Markdown syntax.

140 All generated content and code modifications were manually reviewed and validated by the
141 authors.

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